NAME:_____ Time Start:_____ Time End:_____

Class/Team:_____

Question 9

Show your workings clearly and round off your answers to ***two decimal places*** ***where necessary.*** *Use ^ symbol to represent the power notation.*

At a self-service ordering kiosk, an average of 15 customers arrive every hour. The time each customer takes to place their order at the kiosk is a mean of 3 minutes with a standard deviation of 1 minute.

Question 9a (3 marks)

Using Kendall's notation, state a suitable queuing model for the above queuing system. List 2 assumptions of the queuing model.

M/G/1

Assumptions:

1. **Poisson Arrivals:** Customer arrivals occur randomly and independently according to a Poisson process.
2. **FIFO Discipline:** Customers are served on a First-In-First-Out basis.
3. **Infinite Queue Capacity:** There is no limit to the number of customers who can wait in line.

Question 9b (4 marks)

Calculate:
i. Mean arrival rate of customers in per minute (2 marks)
ii. Utilization of the kiosk (2 marks)

Given:

- Arrival rate per hour (λ_{hour}) = 15 customers/hour
- Mean service time ($1/\mu$) = 3 minutes

(i) Mean arrival rate of customers in per minute (λ)

$$\lambda = \frac{15 \text{ customers}}{60 \text{ minutes}}$$

$$\lambda = \mathbf{0.25} \text{ customers/minute}$$

(ii) Utilization of the kiosk (ρ)

$$\rho = \lambda \times \text{Mean Service Time}$$

$$\rho = 0.25 \times 3$$

$$\rho = \mathbf{0.75} \text{ (or } 75\%)$$

NAME:_____ Time Start:_____ Time End:_____

Class/Team:_____

Question 9c (4 marks)

Calculate:
i. The average number of customers waiting at the kiosk, Lq. (2 marks)
ii. The average time customers spend in the system, Ws in minutes. (2 marks)

Given:

- $\lambda = 0.25$
- Mean service time $(1/\mu)$ = 3
- Standard Deviation (σ) = 1
- $\rho = 0.75$

Since this is an **M/G/1** system, we use the Pollaczek-Khintchine formula.

(i) The average number of customers waiting at the kiosk (L_q)

$$L_q = \frac{\lambda^2\sigma^2 + \rho^2}{2(1-\rho)}$$

Substituting the values:

$$L_q = \frac{(0.25^2 \times 1^2) + 0.75^2}{2(1-0.75)}$$

$$L_q = \frac{0.0625 + 0.5625}{2(0.25)}$$

$$L_q = \frac{0.625}{0.5}$$

$$L_q = \mathbf{1.25} \text{ customers}$$

NAME:_____ Time Start:_____ Time End:_____

Class/Team:_____

(ii) The average time customers spend in the system (W_s)

First, find the average waiting time in the queue (W_q) using Little's Law:

$$W_q = \frac{L_q}{\lambda} = \frac{1.25}{0.25} = 5 \text{ minutes}$$

Now, calculate total time in system (W_s):

$$W_s = W_q + \text{Mean Service Time}$$

$$W_s = 5 + 3$$

$$W_s = \mathbf{8.00} \text{ minutes}$$

Question 9d (3 marks)

Suppose the number of customer arrivals has increased to 90 per hour. Will the queuing system still be in steady state? Determine the minimum number of kiosks required if the mean service time remains at 3 minutes.

Given:

- New arrival rate = 90 per hour
- Service rate capacity per kiosk = $60 \text{ mins}/3 \text{ mins per cust} = 20$ customers per hour

Will the system be in steady state?

- **No.** The arrival rate ($90/hr$) exceeds the single kiosk's service rate ($20/hr$). The utilization would be greater than 1 ($90/20 = 4.5$), meaning the queue would grow infinitely.

Minimum number of kiosks required (k): To be in a steady state, the total service rate must exceed the arrival rate ($k\mu > \lambda$).

$$\text{Traffic Intensity} = \frac{\text{Arrival Rate}}{\text{Service Rate per Server}}$$

$$\text{Traffic Intensity} = \frac{90}{20} = 4.5$$

Since the number of kiosks must be an integer greater than the traffic intensity:

$$k > 4.5$$

Minimum kiosks required = 5

NAME:_____ Time Start:_____ Time End:_____

Class/Team:_____

Question 9e (2 marks)

State the following characteristics (assumptions) in order that a M/M/k queuing model can be applied to study a queuing system with multiple kiosks.

i. Service time distribution _____

ii. Single queue or multiple queue lines _____

(i) **Service time distribution: Exponential** (or Negative Exponential) Distribution. *(Note: The second 'M' in M/M/k stands for Markovian service times, which implies an exponential distribution).*

(ii) **Single queue or multiple queue lines:Single queue** (One common line feeding into multiple servers).

Question 9f (2 marks)

Given that the average number of customers in queue in the M/M/k model, Lq is 6.86, calculate the average waiting time of customers in the queue, Wq in minutes (assume arrival rate of 90 per hour).

- $L_q = 6.86$ customers
- Arrival rate (λ) = $90 \text{ per hour} = 1.5 \text{ customers/minute}$

Calculate average waiting time in queue (W_q): Using Little's Law:

$$W_q = \frac{L_q}{\lambda}$$

$$W_q = \frac{6.86}{1.5}$$

$$W_q = 4.5733...$$

Rounding to two decimal places: W_q **= 4.57 minutes**